# Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

## Constant acceleration

| | |
|---|---|
| **Meaning in my words**<br>How would you explain constant acceleration to a friend? | **How can you tell?**<br>The velocities are 1, 3 and 5 m/s one second apart. What repeats? |
| **Example or sketch**<br>If those gains continue, what velocity would you write for the next second? | **Non-example and why**<br>Someone adds 2 m to position every second and calls that speeding up. What pattern did they draw? |

## Known quantities

| | |
|---|---|
| **Meaning in my words**<br>What do we mean by the known quantities in a problem? | **How can you tell?**<br>A cart starts at rest and is timed for 3 s. Which two values can you write down? |
| **Example or sketch**<br>A sensor reads +4 m at the start. How would you enter that known value? | **Non-example and why**<br>A friend lists a guessed final speed among the given values. Where should it go? |

# Frayer models

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## Displacement

| | |
|---|---|
| **Meaning in my words**<br>How would you explain displacement to a friend? | **How can you tell?**<br>Which two marks tell you the displacement even if the cart turns around? |
| **Example or sketch**<br>Go from 2 m to 6 m and back to 4 m. What single arrow shows the displacement? | **Non-example and why**<br>Someone adds both legs and gets 6 m. What did they find? |

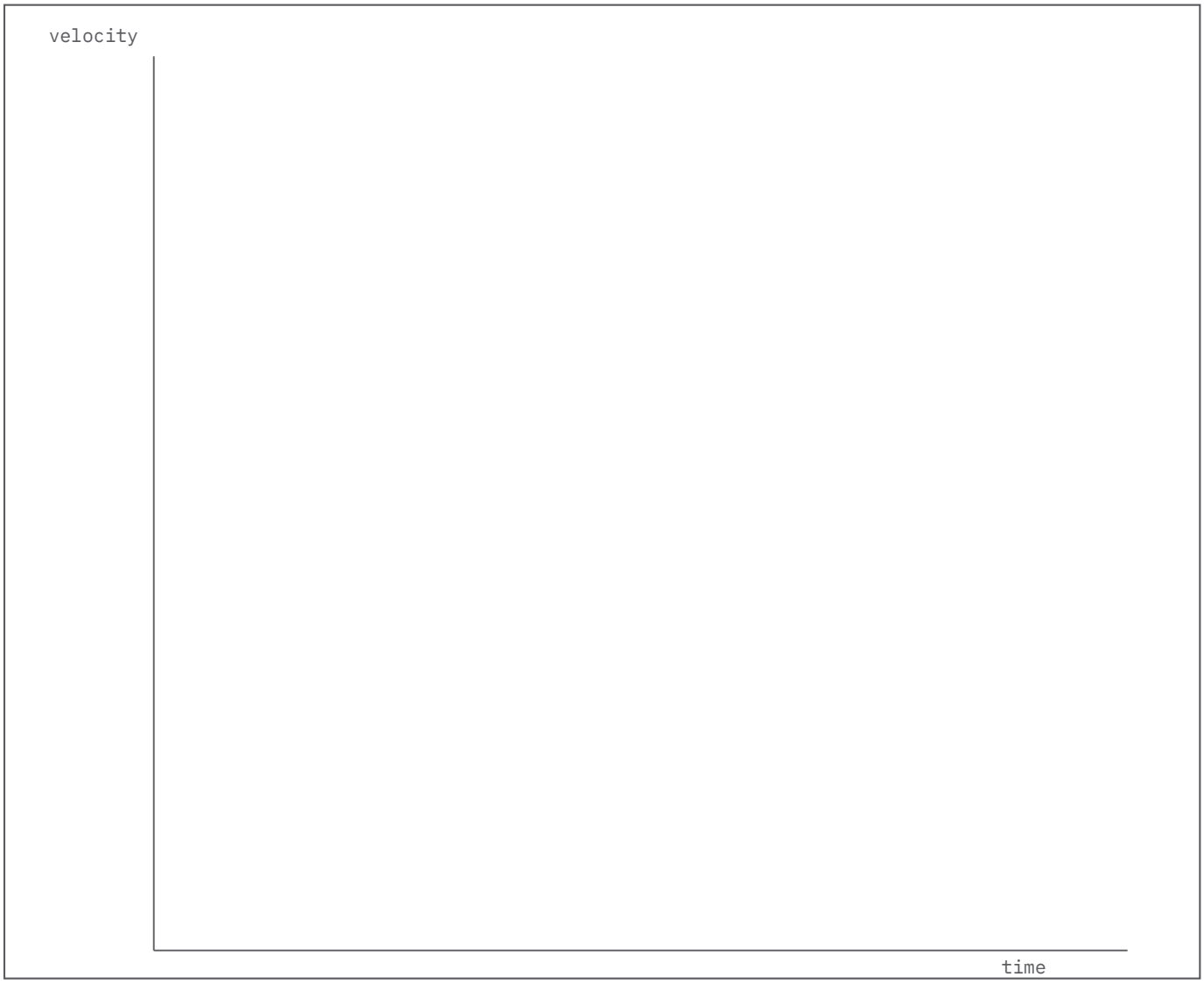
## Cross-check

| | |
|---|---|
| **Meaning in my words**<br>How would you explain cross-check to a friend? | **How can you tell?**<br>You found displacement using an equation. What feature of a velocity–time graph could check it? |
| **Example or sketch**<br>A cart stays at +3 m/s for 2 s. How can a graph check the result of 6 m? | **Non-example and why**<br>Someone copies the same number twice. What could actually catch a wrong answer? |

# Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

## Sketch a straight velocity–time line. Label which feature gives acceleration and which gives displacement.




**My marks mean**

# On assignment: Constant acceleration

SECTION E / BACK / FINAL MEETING ONLY Name _____

**XJ moves from +2 to +6 km/s over 2 s. The rehearsal assumes constant acceleration. Where should the next camera look relative to the start?**

## PICK A JOB / DELIVER ONE PRODUCT

☐ **Camera engineer**

Draw the velocity–time trace and shade the area. Send the camera its displacement for the end of the shot.

☐ **Replay editor**

Sketch the opening and closing speeds and give a voice-over explaining why using the final speed for the whole clip would miss the shot.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

## INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor's check: Cross-check with a kinematics equation and name what it assumes.